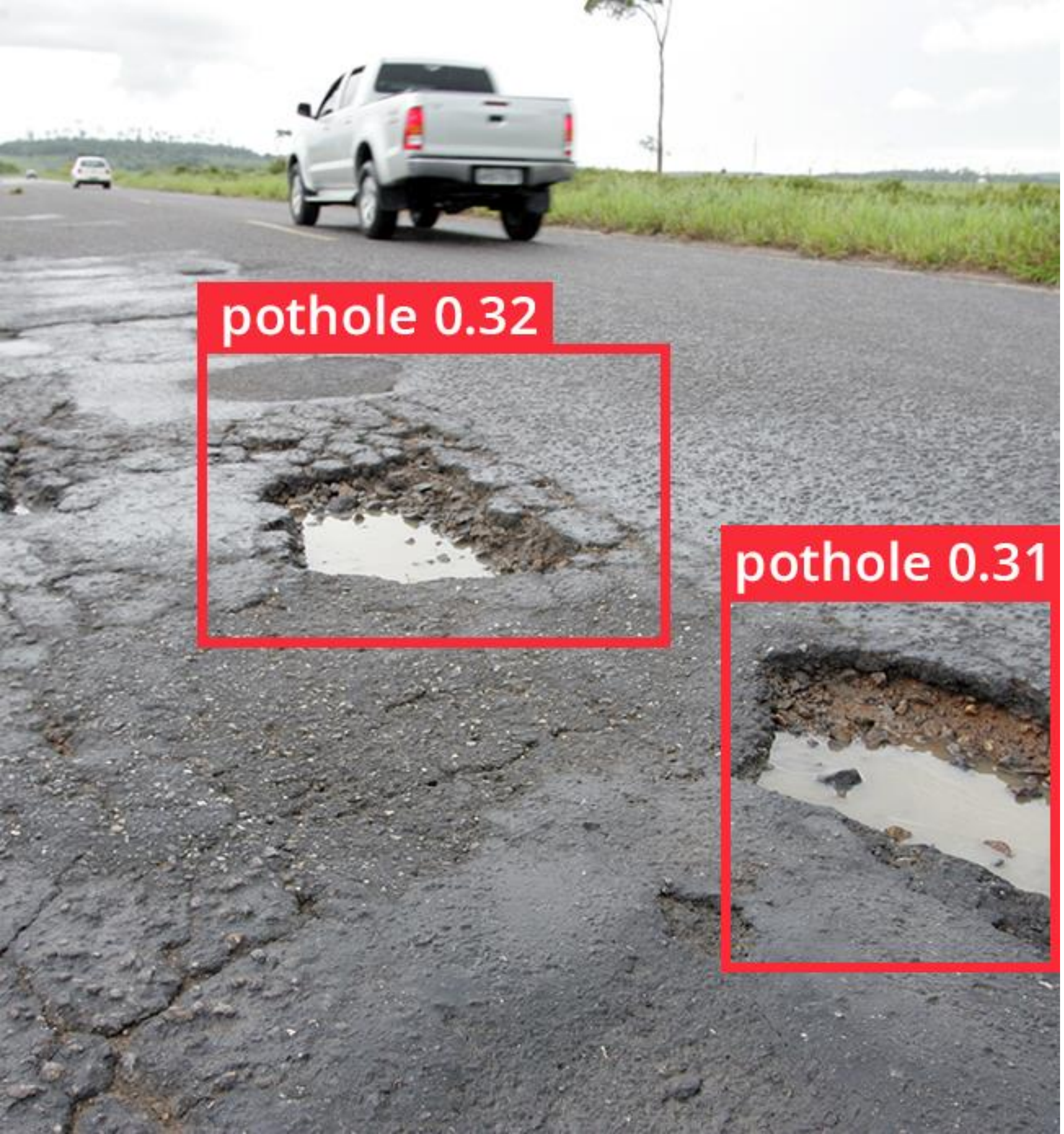


Detection of Potholes around Bangor University premises from Traffic Videos using Computer Vision Techniques

By: Ubong Camilus Ben

PROBLEM STATEMENT

Detection and estimation of pothole dimensions play a vital role in effective road maintenance. Potholes commonly form due to aging pavement, heavy rainfall, continuous traffic load, and weak underlying layers.

These potholes pose serious safety risks. Drivers may lose control when hitting or trying to avoid them, potentially leading to injuries or even fatal accidents. Additionally, potholes often cause property damage, such as flat tires, vehicle scrapes, dents, and fluid leaks.

Traditionally, pothole assessment involves manual field surveys conducted by trained personnel to estimate their size for planning repairs and calculating costs. However, this method is expensive, time-consuming, and often subjective. It can also be unsafe and prone to human error.

To address these limitations, this project proposes a modern, real-time, and cost-effective solution. A deep learning-based method will be developed to detect potholes directly from real-time video footage, offering greater accuracy, efficiency, and safety in road maintenance operations

LITERATURE REVIEW

- Numerous methods exist in the literature for the detection and size estimation of potholes.
- Pena-Caballero et al. (2020) compared the performance of object identification and semantic segmentation algorithms with regard to detection time and overall system accuracy.
- Park et al. (2021) used YOLOv4, YOLOv4-tiny, and YOLOv5s for pothole detection. Compared to YOLOv5s, both YOLOv4 and YOLOv4 Tiny performed marginally better, with respective accuracy levels of 77.7% and 78.7%.
- Hedeya et al., (2022) used 3D point clouds to classify the severity of potholes based on region of interest (RoI) extraction.
- Mask Region-Based Convolutional Neural Network (Mask R-CNN) was also used by Arjapure and Kalbande (2021), who employed the approach to detect potholes from 2D images.
- Most studies focused on images with characteristic paucity of studies using videos for pothole detection.

DATA SOURCE AND METHODOLOGY

- Data type:

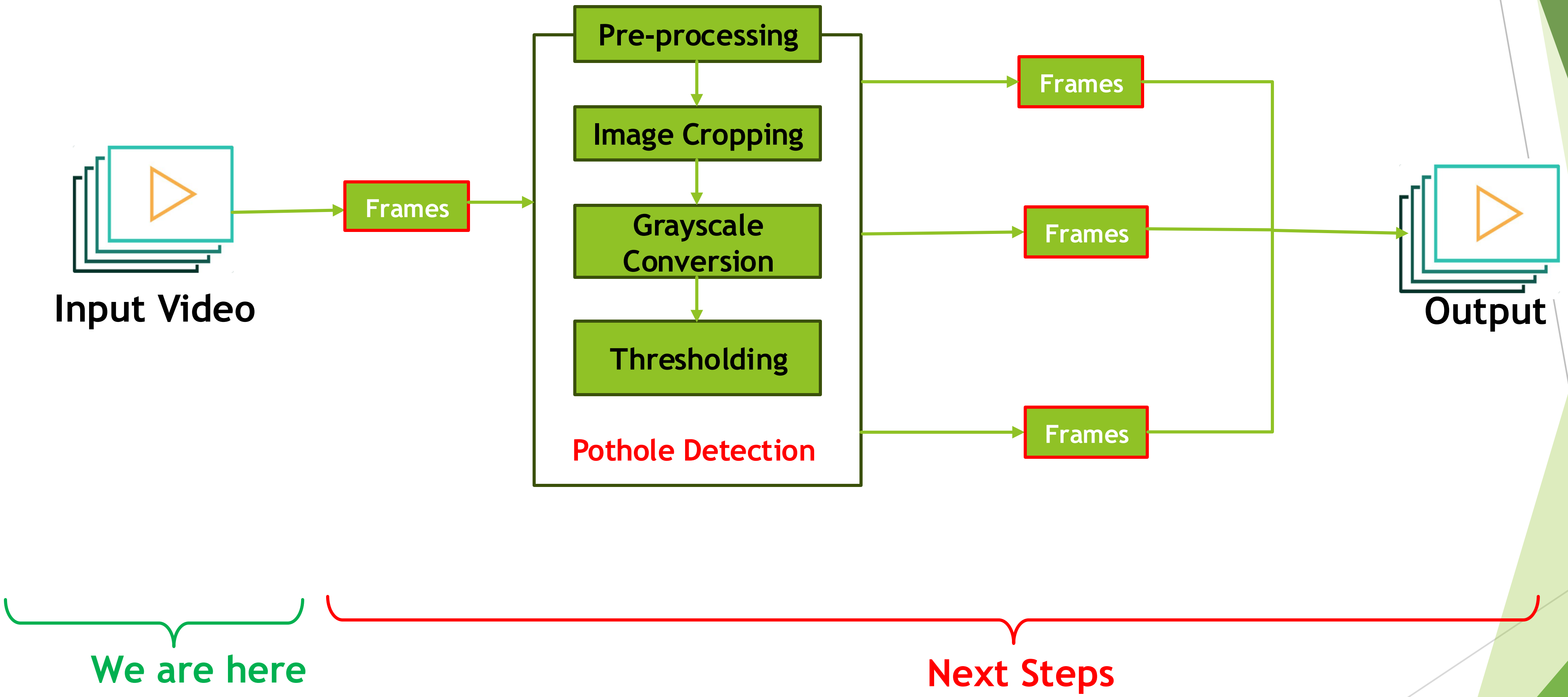
Annotated pothole images and dashcam videos.

- Data Source:

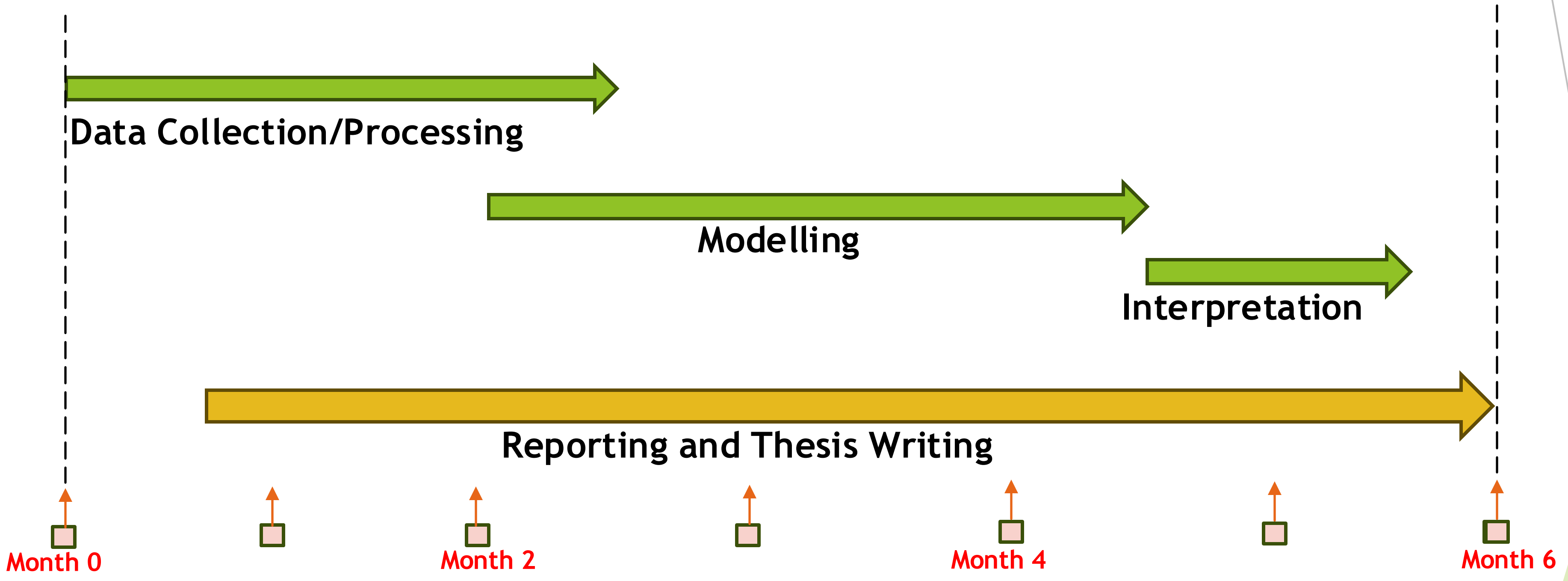
Scopus Data Repository: https://data.mendeley.com/datasets/tp95cdvgm8/1/files/ffad1712-984b-4b70-bb80-39697045bd9e

The samples of road damage were collected from different sections of road that experienced road damage in industrial districts around the UK. All images have resolutions of resolution of 1920 × 1080 pixels

PLANNED APPROACH



TENTATIVE DURATION



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